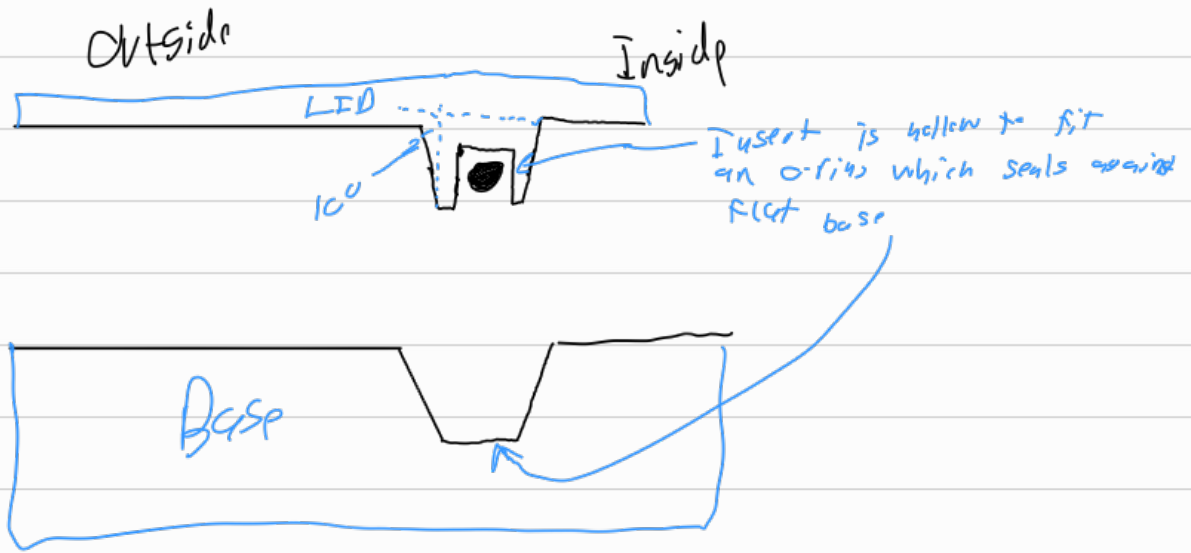
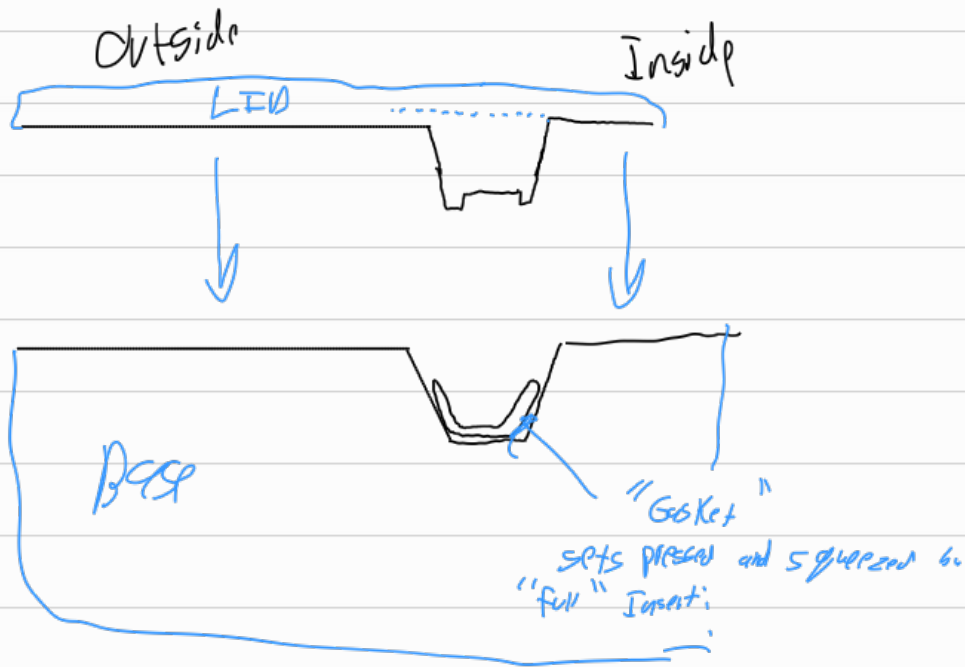


Seals:

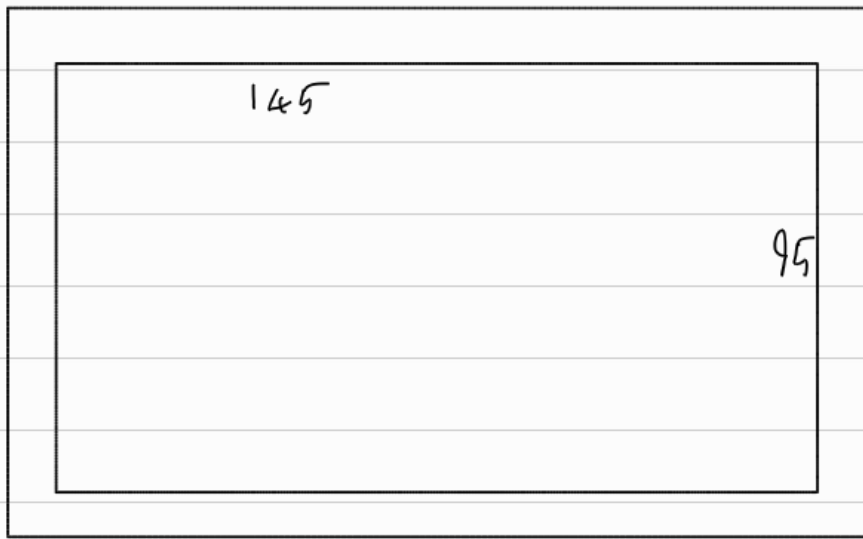
Option 1: O-rings smooth



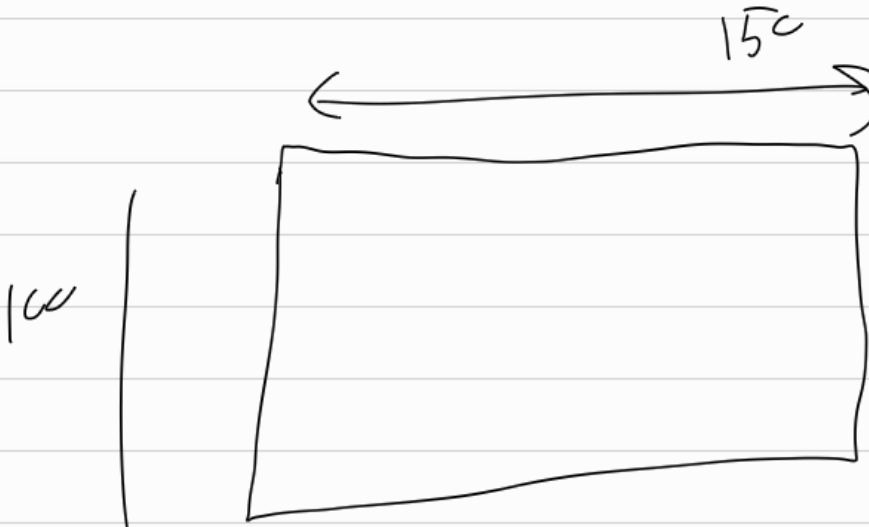
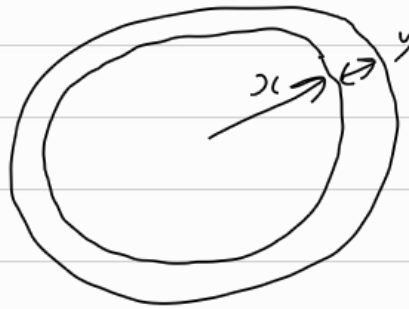
Option 2: Pliable Gasket



$$\frac{\text{Perimeter} - 100}{4} + 50$$



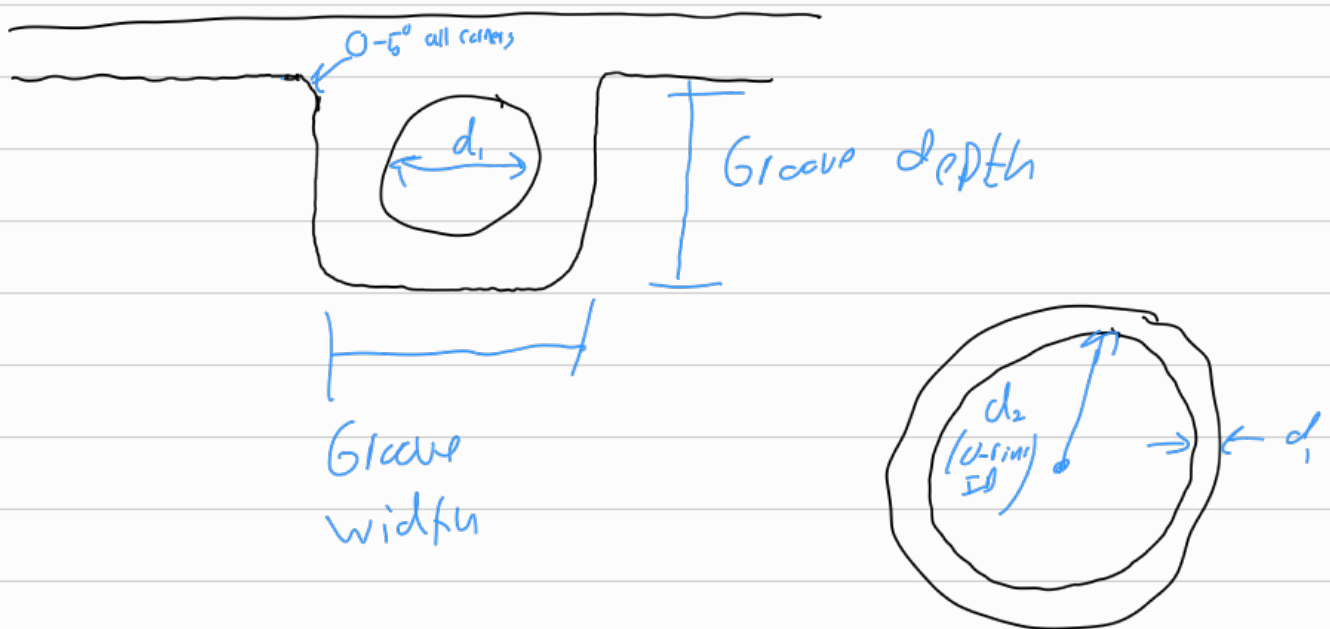
$$\frac{\text{Perimeter} - 100}{4}$$



$$1.5 : 1$$

900

O-Ring Groove design



- For perimeter, want minimum O-ring stretch $\therefore$ Mid line of groove Diameter = middle circumference of O-ring
- Groove width should allow max 5% expansion of $d_1 \therefore Gw = 1.05d_1$
- Groove depth: for fire seal, Ideal compression ratio between 20-30%

$$\text{Compression} = 0.2 = \frac{d_1 - \text{Groove depth}}{d_1} \quad \begin{aligned} 0.2d_1 &= d_1 - Gd \\ Gd &= d_1 - 0.2d_1 \\ &= 0.8d_1 \end{aligned}$$

$$\therefore \text{Groove depth} = d_1(1 - 0.2) = 0.8d_1$$

- For rectangular groove: corners must be rounded to reduce stress in O-ring.
- $\therefore$ Add Fillet on corner, where Inner Fillet radius $> 3X$ (6X) O-ring cross sectional diameter.

LID : data dim = $160 \times 110 \times 5$ mm

- 2 versions, 1 for O-ring, 1 for Gasket
- uses 4 Max bolts for securing (can get rid of for latch)
- Inside of Sealing feature is raised slightly higher than outer level, this lets outer level sit flush with base.
- Inner Panel groove has 1.1 mill indent for mounting panel.

Base : • has Groove to accept Sealing feature on Lid.

• Has $128.8 \times 86.5 \times 20$ mm Chamber for Multi-well plate

• Has Simple Face Seal O-ring dosion where plate is pressed against the C-ring to get Seal

↳ Plate pressed from top to get force for seal

• Has 108.8×66.5 mm Viewing opening.

- O-ring lock
- rounded corners
- bottom seal
- Indents for fasteners
- Check printing recs

look at heat inserts

→ top fast (fasteners) (look for standard sizes)

→ (cast) seals

127.4

124.5

85.5